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ATTACHMENT, LOSS, AND DEPRESSION*

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INTRODUCTION

THE DISRUPTION of an attachment bond is a traumatic occurrence for higher primates. Human infants deprived of proper mothering may develop the syndrome of anaclitic depression (Spitz, 1946); and children who have been separated from their mothers often exhibit an initial period of protest, followed by a behavioral reaction which Bowlby (1952, 1961) described as "despair". In adults, grief and bereavement are accompanied by increases in both mortality and morbidity among the survivors (Black, 1978; Epstein, 1975; Jacobs and Ostfeld, 1977; Parkes, 1972; Rees and Lutkins, 1967), and an increased incidence of juvenile rheumatoid arthritis has been reported in children who have experienced a major loss (Henoch, Batson and Baum, 1978). The relationship between separation, loss and depression, while moot, is hardly inconsequential (Cadoret *et al.*, 1972; Heinicke, 1973; Paykel, 1975; Stenback, 1965; Sethi, 1964). This is perhaps most especially true for children. In a study of 50 children with affective disorders, McKnew and Cytryn (1973) stated: "In summary, our investigation suggests that the *sine qua non* in acute depressive reactions is the sudden loss of a love object . . .".

Profound behavioral reactions, with a depressive affective component, are also found in young non-human primates following loss or separation. In chimpanzee infants the loss of the mother may result in the death of the orphaned infant, even though it is adopted and cared for by others (Goodall, 1971a, 1971b, 1979). In monkey infants, separation from the mother or from peers (and in monkey adults separation from the nuclear family) results in an agitation–depression or protest–despair reaction that may have an associated mortality, and may, in cases of repeated peer separations, result in an arrest of growth (Harlow and Suomi, 1974; Kaufman and Rosenblum, 1967a, 1967b; Hinde, Leighton-Shapiro and McGinnis, 1978; Hinde and Spencer-Booth, 1970; Seay, Hansen and Harlow, 1962; Suomi *et al.*, 1975; Soumi *et al.*, 1976; Mineka and Suomi, 1978).

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Attachment, separation, and loss are areas of vital importance to psychiatry. In fact, however, we know very little of the physiological mechanisms which underlie the response to loss or separation, mechanisms which account for affected individuals being more susceptible to a variety of insults. From a clinical standpoint, an understanding of such mechanisms is important both because of its relationship to the pathophysiology of grief, and to the etiology of affective disorders.

Studies in our laboratory over the last several years have been designed to investigate the physiological sequelae of the response to maternal separation in infant pigtailed (*M. nemestrina*) monkeys. In previous papers we have presented data on physiological correlates of 4 days of maternal separation in unrestrained group living pigtailed infants (Reite *et al.*, 1978; Reite and Short, 1978; Seiler *et al.*, 1979). In this paper we present more extensive physiological and behavioral data on the effects of a 10-day maternal separation in a group of 8 pigtailed infants. Our goal in presenting these new data is threefold. First, we think it important to replicate our original findings in this new group of animals. Secondly, an examination of the longer 10-day period of separation permits us to examine the evolution of the depressive reaction in more detail, including the evaluation of possible early behavioral and/or physiological recovery. Thirdly, since early developmental data is available for the present group of animals, we can evaluate to what extent individual differences in the response to separation may be predicted by measures of early behavioral development and the early mother–infant interaction.

The mother–infant bond is the prototype of much in the way of future attachment bonds and other forms of affiliative behavior, and there is good reason to believe that the response to the disruption of the mother–infant attachment bond is homologous (e.g. sharing both common functions and underlying mechanisms due to a common evolutionary history) among the high primates (Bowlby, 1969; Reite, 1977). Thus, what we learn from a monkey model may have general relevance to our understanding of man.

METHODS

The pigtailed infants used in this study (see Table 1) were born to wild-born mothers in the Department of Psychiatry Primate Laboratory during 1975 and 1976. All experimental animals were raised by their mothers in social groups containing one male and four to eight adult females, some with infants.

TABLE 1. A LIST OF THE INFANTS DESCRIBED IN THIS PAPER

Animal no.	D.O.B. day/yr	Sex	Age at implantation	Age at separation
R8.5	192–76	F	118	148
R9.3	21–76	M	126	172
R11.5	100–76	M	152	176
R20.2	136–76	F	143	168
R22.1	82–76	M	119	146
R23.1	229–75	M	128	152
R23.2	243–76	M	108	131
R25.1	218–75	F	182	205

D.O.B. = date of birth in day of year and year.
All ages given in days.

The groups were housed in group pens measuring 2.1 × 2.5 × 4.0 m with glazed cinderblock walls, wire mesh ceilings, and pipes and shelves positioned on one or more walls. One or two walls contained one-way windows for observational purposes. The groups were fed at 0900 daily, and water was available *ad lib*. Timer controlled fluorescent lights were turned off between 2000 and 0700 hours.

Starting with the first month of life, periodic behavioral observations were made on all infants. A trained observer watched the animals through one-way windows for four 3-min sessions 3 days a week, and dictated observed behavior onto an audio-cassette. The cassette was later replayed while the observer listened to it through earphones and entered the specific behaviors as they occurred into a PDP-12 computer keyboard. The computer provided duration (in seconds) and frequency scores of all behaviors observed during the session (Walker and Reite, 1975). Behavioral data collected during the first 4 months of life was collapsed across 28-day (4-week) months and each variable was converted to a duration (percentage of total time observed or % TTO) or frequency (counts/1000 sec observed) score. Specific behaviors discussed in this paper are listed in Table 2. For ease of presentation, the different

TABLE 2. A LIST OF BEHAVIORAL VARIABLES DISCUSSED IN THIS PAPER

Group	Behaviors	Type	Description
Activity	Activity Count	F*	A measure of the infant's activity (Reite and Short, 1980)
	Motion	D†	Locomotion—walking, running, or climbing
	Rest	D	At rest—no visible movement. Eyes may be open or closed
	Slouch	D	Characteristic posture of depression in which the the infant sits hunched over, often with the head down, and yet is awake
Social Contact	Play	D	Social play (with other animals)
	Contact Other	D	Physical contact with another animal (other than mother)
	Initiate Groom	D	Grooming another animal
	Receive Groom	D	Receives groom from another animal
	Receive Threat	F	Receives a threat from another animal
Environmental Contact	Oral Obx	D	Exploration of inanimate object with mouth
	Obx	D	Visual examination of inanimate objects usually while manually manipulating them
	Obs	D	Exploring sawdust or woodchip bedding on pen floor
Self Directed	Oral Self	D	Self-mouthing and self-licking
	Chew	D	Chewing of foodstuffs, e.g. biscuits, fruit
	Auto Groom	D	Animal grooms itself
	Auto Play	D	Playing alone
	Auto Gen	D	Self directed genital exploration
Mother–Infant	Physical Contact	D	All times the infant was physically in contact with the mother
	Off	D	All times the infant was not in physical contact with the mother
	Other Level	D	The infant is on a different vertical level from the mother
	Away	D	All times the infant was on another level from the mother and/or more than 60 cm from her
	Nipple	D	Infant is holding the mother's nipple in its mouth
	Wean	F	Nipple withdrawal. Mother removes nipple from infant's mouth
	Punish	F	Infant is punished by mother or other adult

*Frequency items.
†Duration items.

behavioral variables have been classified into one of 5 major groups (Activity, Social Contact, Environmental Contact, Self Directed, and Mother–Infant).

Additionally, whenever the distance between mother and infant increased to more than 60 cm (scored as a "leave"), or decreased from greater than 60 cm (scored as an "approach"), a record was made of who initiated the leave or approach. If the mother left the infant, a score of "mother-leave" (L_m) was obtained; if the infant left the mother a score of "infant-leave" (L_i) was obtained, and similar scores were obtained for approaches (A_m and A_i). Using the leave and approach scores, we calculated the differences between the percentage of approaches initiated by the infant and the percentage of leaves initiated by the infant using the following formula after Hinde and Atkinson (1970):

$$\%A - \%L = \frac{A_i \times 100}{A_m + A_i} - \frac{L_i \times 100}{L_m + L_i}$$

If this value is negative, it suggests that the infant is initiating more of the leaves (which means the mother is primarily reponsible for maintaining proximity), and if positive, the infant is initiating more approaches (and is therefore primarily responsible for maintaining proximity).

At a mean age of 19 weeks (range 15–26 weeks), the infants were surgically implanted with multichannel biotelemetry systems which enabled the recording of EKG, body temperature (BT), eye movement (EOG), muscle activity (EMG), and 3 channels of EEG from the unrestrained infant living in its social group (Reite, Walker and Pauley, 1973; Pauley, Reite and Walker, 1974; Reite and Pauley, 1976). Following recovery from surgery the infants and their mothers were returned to the social groups, and physiological data recordings began shortly thereafter (Reite *et al.*, 1974b). Physiological variables to be discussed in this paper are listed in Table 3. Details of the methods used to score and analyze the physiological data are described in Reite and Short (1980). While physiological data was being recorded, behavioral observation sessions were increased to six 5-min sessions a day.

TABLE 3. DEFINITIONS OF PHYSIOLOGICAL VARIABLES

Variable	Brief description
Mean day HR	The mean of all heart rate samples obtained between 1000 and 1600 hours
Mean night HR	The mean of all heart rate samples obtained between 2200 and 0400 hours
Mean day BT	The mean of all body temperature samples obtained between 1000 and 1600 hours
Mean night BT	The mean of all body temperature samples obtained between 2200 and 0400 hours
Sleep latency	Time between lights out and sleep onset
Awake	Time awake after sleep onset and before final arousal preceding lights on in the morning
No. arousals	Number of awakenings between sleep onset and final morning arousal
Total sleep time (TST)	Minutes of total sleep (a combination of Drowsy, Stage 1, Stage 3 + 4, REM)
Drowsy	See Reite *et al.*, 1974 for description
Stage 2	Stage 2 sleep
Stage 3–4	Stages 3 and 4 are combined
Slow wave sleep (SWS)	Drowsy, Stage 2, and Stages 3–4 combined
REM time	Minutes of REM sleep
No. REM Pds.	Number of REM periods during the night
Mean REM length	Mean length of REM periods for a given night
REM latency	Time from sleep onset to beginning of first REM period
IRI	Mean interval between REM period onset times
Alpha frequency	Mean frequency of EEG alpha activity as determined from power spectral analysis (see Reite and Short, 1980, for details)
Alpha ratio	Ratio of the EEG spectral power in the 6–10 Hz frequency band divided by the power in the 3–6 Hz band (see Reite and Short, 1980, for discussion)

After a three to five day period of preseparation (baseline) physiological and behavioral data recording, the mothers were separated from the infants. The mothers were housed in individual cages in another part of the laboratory for the duration of separation whereas the infants remained in their social groups. Separations normally took place between 1300 and 1400 hours. Physiological and behavioral data collection continued for the duration of the separation. After 10 days of separation, the mothers were returned to the social groups, again between 1300 and 1400 hours. Physiological and behavioral data continued for another 4 days (reunion).

For purposes of data analysis, mean scores were computed for each infant for each behavioral and physiological variable during the baseline (*B*) period, and for the subsequent separation (*S*) and reunion (*R*) periods, which were divided as follows: S_A includes all data collected from the moment of maternal separation through the first night of separation. S_B includes the second, third and fourth days and nights of separation. S_C includes the fifth through seventh days and nights of separation, and S_D includes the eighth through tenth days and nights of separation and first half of the eleventh day of separation (that time prior to reunion). R_A includes all data collected from the moment of reunion through the first night of reunion, and R_B includes all subsequent reunion data (second through fourth days and nights). Technical considerations precluded the collection of all physiological data from all animals.

Most behavioral and physiological variables will be discussed in terms of their percentage change (% change) from the baseline group mean values. Certain behavioral variables (Slouch and Auto Gen) have a near zero mean score during baseline, hence separation scores will be presented in terms of actual percentage of total time observed (%TTO) scores. To obtain the percentage change scores, we subtracted each infant's baseline mean score from its 4 separation (S_A through S_D) or reunion (R_A and R_B) period scores, and converted the result to its percentage of the baseline mean.

All behavioral and physiological data were processed using the Statistical Package for the Social Sciences (Nie *et al.*, 1975). Each condition mean for each variable (% change) along with the 95% confidence intervals are used in most illustrations. The Spearman rank order correlation matrix was examined to assess possible relations between variables for each condition. When examining the maintenance of rank across conditions for the physiological and behavioral variables and the patterns of correlation from condition to condition, collected scores were used; when examining the relationships between developmental behavior and changes in physiology, the percentage change from baseline scores was computed, ranked, and the correlated with the ranks found for behavioral data.

RESULTS

1. *Behavior*

Our infants vigorously protested separation from the mother, with much cooing (the distress call of the young Macaque) and screeching. Activity Count and Motion (Fig. 1) showed marked increases immediately following separation (S_A day) coinciding with the agitation reaction. Rest decreased markedly; in fact, no infant rested after separation until after lights-out that night. During the remainder of separation, Activity Count was slightly depressed, particularly during separation days 5–7 (S_C), although Motion scores remained about the same as baseline. Rest behavior, on the other hand, remained lower than baseline throughout separation. These quantitative scores do not reflect the qualitative changes in the locomotion of the depressed infants who would often appear to locomote as if in slow motion and who also frequently evidenced impaired motor coordination. Depressed infants could, however, at times of maximal arousal (during feeding and at times of social group stress) marshall their resources and move with normal speed.

Immediately following reunion with mother (R_A), the infant's Activity and Motion scores were very low, and rest scores increased; the infant was spending most of its time in physical contact with the mother. During the remainder of reunion (R_B) Rest, Motion, and Activity Count scores returned toward baseline values.

The slouched posture of depression (Slouch, Fig. 1) was not usually observed until the second day of separation. Slouch scores for the group were highest during the S_C period, with a return to baseline (essentially zero) values following reunion. The slouched posture was also frequently accompanied by a characteristic facial expression similar to that described by Darwin (1872) as ". . . universally and instantly recognized as that of grief . . .".

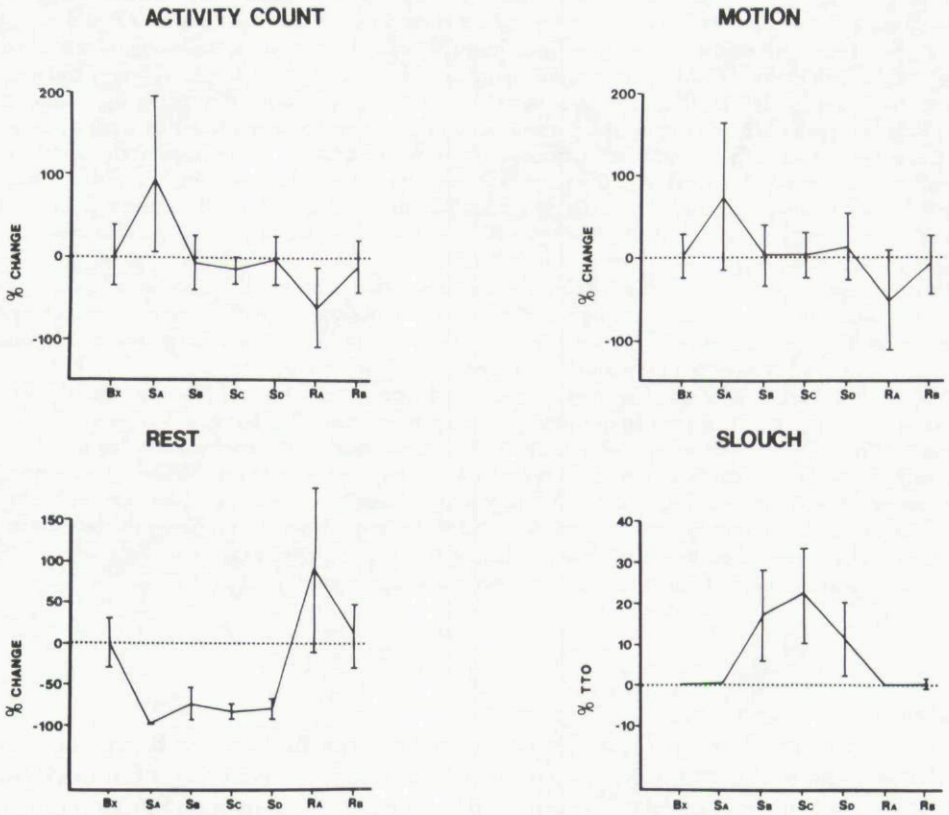


FIG. 1. Changes in scores for activity group variables during separation (S_A through S_D) and reunion (R_A and R_B). Baseline mean values (B_X) are represented by the dotted lines. On this and subsequent illustrations, the vertical brackets for separation and reunion represent 95% confidence intervals for change from baseline in percent. For B_X the brackets indicate the 95% confidence interval across animals converted to % of baseline.

Among the Social Contact category behavioral variables (Fig. 2), a decrease in Play behavior was seen in all infants during separation, and the separated infants no longer responded to Play overtures initiated by other infants. There was a tendency for Play scores to increase slightly as separation continued, suggesting a slow recovery function. Following reunion, Play decreased the first day (the infant spent most of its time with the mother), and then returned toward baseline values. The decrease in Play behavior was not a manifestation of a more general social withdrawal, however, since separated infants frequently attempted to make contact with other (usually

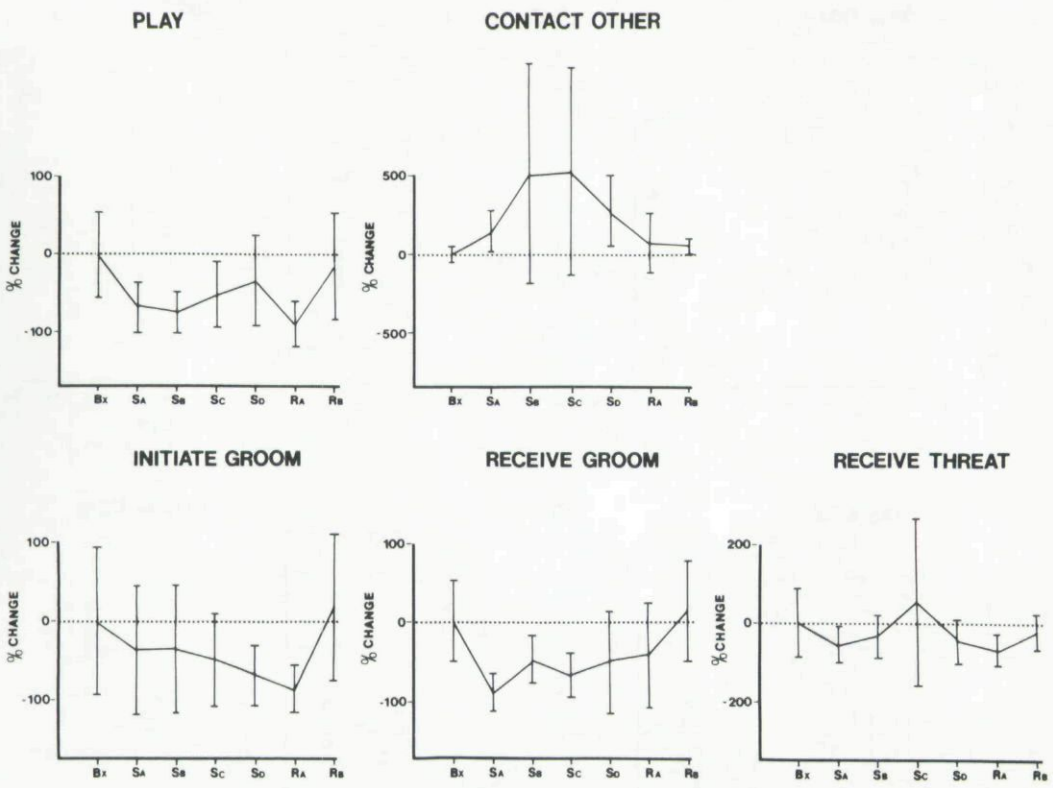


FIG. 2. Changes in social contact group behaviors during separation and reunions. Legend as per Fig. 1.

adult female) animals; this tendency is illustrated in the Contact Other scores, which increased markedly during separation (Fig. 2). The amount of grooming initiated and received by the separated infant decreased during separation (returning to normal following reunion), in spite of the increase in Contact Other behavior. Grooming *per se* is rather infrequent in this age range (Defler, 1978; Reite and Short, 1980), and most of it is directed to or received from the mother. Self-grooming (Auto-Groom) behavior did increase, however, as did other self-directed behaviors discussed below.

As a rule, the separated infants were not threatened more often by other group members (Receive Threat, Fig. 2) while the mother was gone, although there were isolated cases of separated infants being aggressed against by adults. In no case did such an occurrence result in injury, however.

Among the environmental contact category behaviors (Fig. 3), increases were seen in all three behaviors during separation, with large increases in Oral Obx appearing shortly after separation (during the S_A day). All three behaviors returned to values lower than baseline on the first (R_A) day of reunion with the mother.

Self-directed category behaviors (Fig. 4), with the exception of Auto Play, generally increased through separation. This was especially true of Chew behavior which, although it did not increase immediately following separation (S_A day), had values

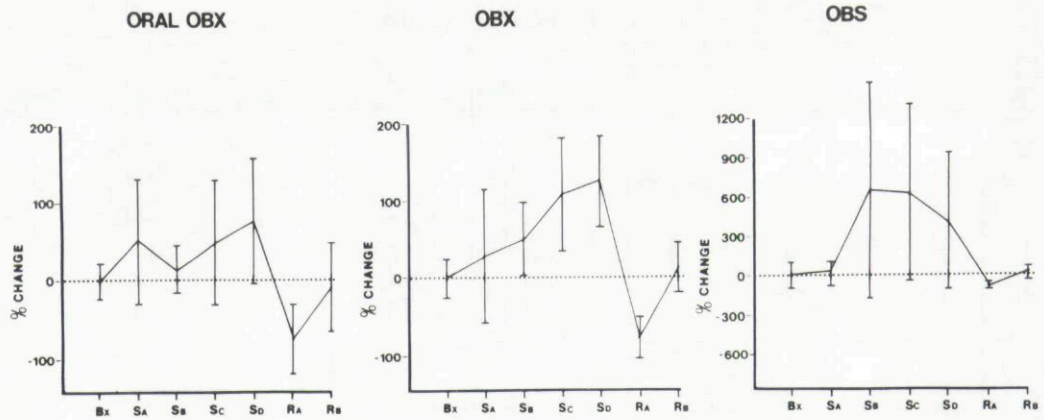


FIG. 3. Changes in environmental contact behaviors during separation and reunion. Legend as per Fig. 1.

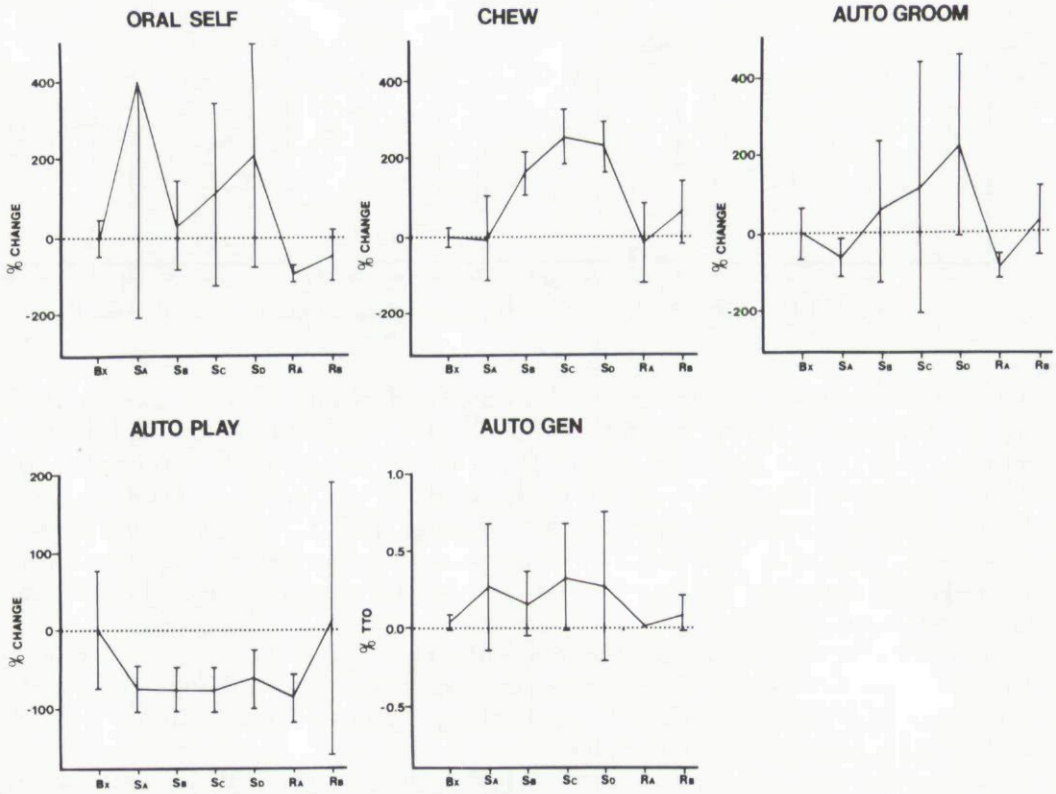


FIG. 4. Changes in self-directed behaviors during separation and reunion. Legend as per Fig. 1.

of about twice baseline for the remainder of separation. The slight increase in Auto Gen behavior during separation was largely confined to the male infants. Although showing some individual variability, self-directed behaviors generally tended to normalize following reunion with the mother. Self-orality behaviors were most

prominent on the first day of separation when vigorous self-mouthing would often accompany the agitation reaction; these behaviors remained elevated throughout separation. Changes in the mother–infant relationship category behaviors (Fig. 5) during the reunion period in comparison to baseline gives some indication of the effects of the separation experience on the subsequent mother–infant relationship. Physical Contact increased immediately following reunion (R_A day), but the increase was not sustained. In a similar fashion, scores for Other Level and Away decreased, with Other Level scores remaining lower than baseline throughout the R_B period. Even though the infant spent more time in Physical Contact the R_A day, Nipple scores did not increase, probably due to the fact that often the infant would fall asleep very shortly after attaining ventro-ventral contact with the mother. Wean and Punish scores were lower than baseline during the R_A and R_B periods, suggesting that the mother was more tolerant of the infant, or perhaps that the infant was being generally quieter and more subdued (as suggested by the decrease in Activity Count and Motion and increase in Rest scores during reunion illustrated in Fig. 1).

In summary, the infants' reaction to separation was accompanied by quantitative and qualitative changes in the nature of the separated infants' motor activity, a

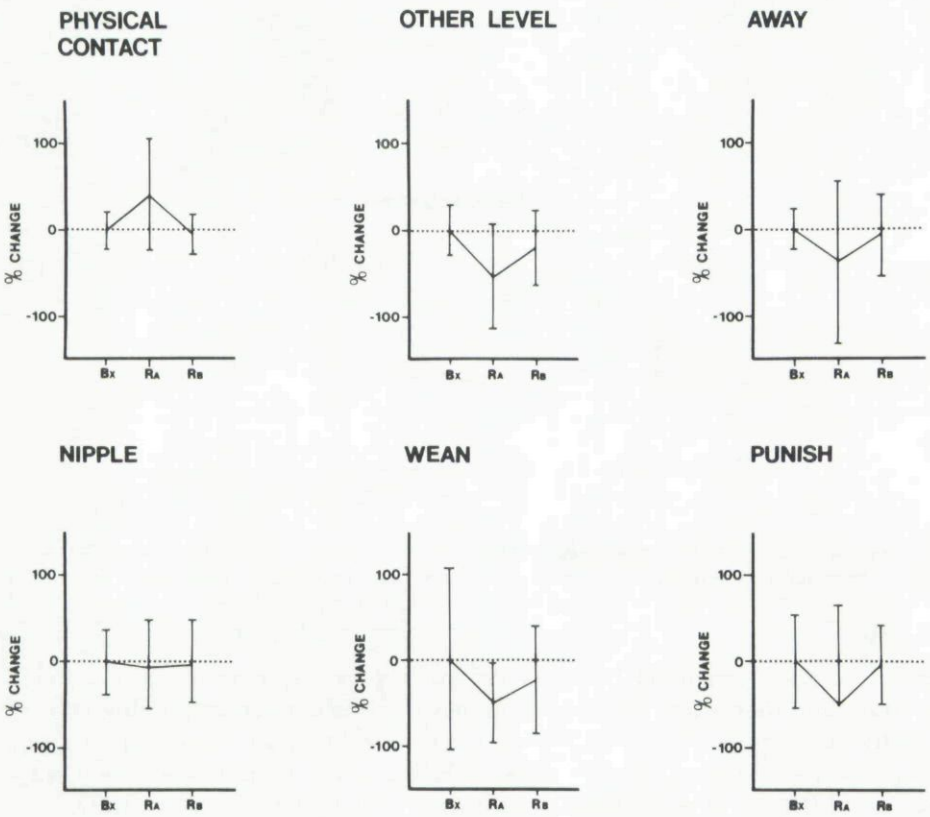


FIG. 5. Changes in mother–infant behaviors during the reunion period following 10 days of separation. Legend as per Fig. 1.

decrease in certain social behaviors, most prominently Play, and a marked increase in self-directed behaviors, especially oral behaviors. This would appear consonant with a withdrawal of interest from the social environment and a redirection of attention to internal stimuli. These behavioral changes were for the most part reversed following reunion with the mother, although the nature of that relationship itself was also influenced by the separation experience.

2. *Physiology*

a. *Heart rate.* Immediately following separation and accompanying the "agitation" reaction, heart rate (HR) increased in all animals. This is illustrated in Fig. 6 where mean day HR for the S_A day was 10% higher than baseline values for the group.

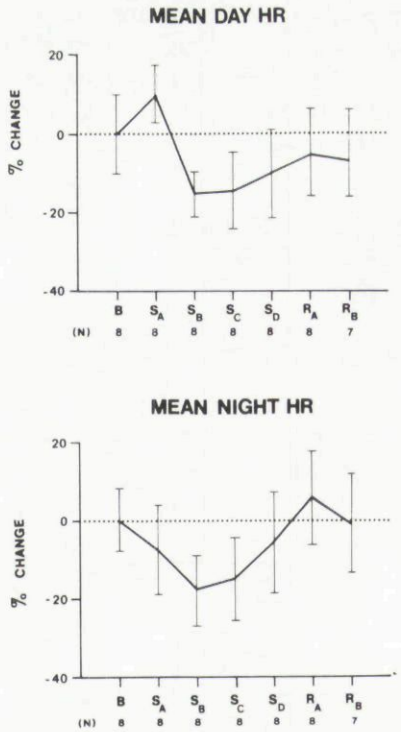


FIG. 6. Per cent change (from baseline) in mean day HR and mean night HR during separation and reunion. The small (N) beneath the experimental time periods indicates the number of animals for whom data was available.

Often the highest single HR values recorded were those immediately following separation, and these were equal to or higher than values accompanying active play during the baseline period (even though the physical activity accompanying separation was not as vigorous). The increase in HR usually did not persist beyond sleep onset the first night of separation, and during that first night of separation, mean HR was lower than any preceding baseline night in 5 of the 8 infants (Fig. 6, mean night HR, S_A).

The lowest mean HR values for the group both day and night were seen during the first third of separation (S_B); a gradual recovery towards baseline values accompanied the remainder of separation (S_C and S_D) with 4 of the 8 infants reaching baseline night values by the 10th night of separation. Individual variability was marked. While most infants (6 of 8) had their lowest mean night HR during nights S_1 to S_4, in one infant (22.1), the lowest mean night HR was the 6th night, and in one (9.3), on the 10th night. We have also found the incidence of cardiac arrhythmias to increase following separation (Seiler *et al.*, 1979), but since arrhythmias data was not obtained throughout separation in the infants described in this paper, we will not consider it further here.

Group mean HR values on the first day of reunion (R_A) were lower than baseline. During the first reunion night, however, HR was higher than baseline in 5 of the 8 infants, with group mean values returning to baseline during the remainder of the reunion. Several infants had heart rates higher than baseline during the remainder of reunion, and 2 infants (9.3 and 23.1) had sustained decreases in HR during the remainder of reunion.

b. *Body temperature.* Body temperature (BT) was recorded from 7 infants and of these 7, BT measures were not obtainable for one animal after the first reunion day. Figure 7 illustrates the mean percent change from baseline for the group in mean day and mean night BT. Mean day BT following separation and during agitation

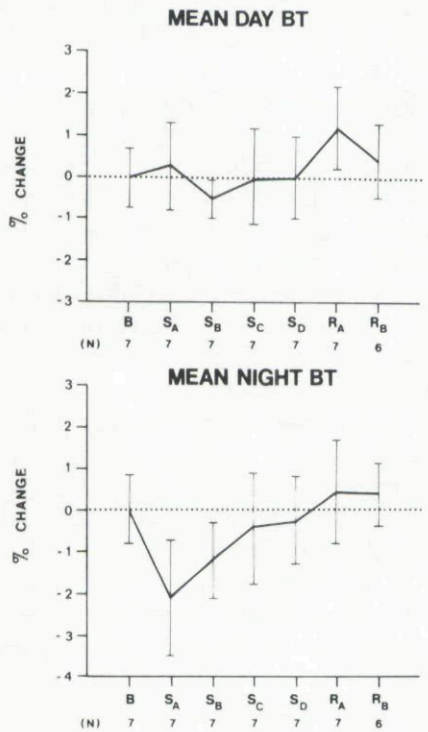


FIG. 7. Per cent changes (from baseline) in mean day BT and mean night BT during separation and reunion.

was greater than any preceding mean day BT value in 6 of the 7 infants. It was striking to observe a recently separated infant sitting on the bar or bench in the group pen—perhaps running about from door to window on occasion (but not as physically active as during vigorous play)—with its BT quickly climbing over several minutes to relatively high values (compared to baseline)—apparently a manifestation of its high level of arousal and a general increased metabolic rate.

The first night of separation, mean night BT decreased to a value lower than any preceding baseline mean night BT value in 5 of the 7 infants. In the other two, it equalled the lowest baseline night value. For the group as a whole, mean night BT values remained lower than baseline only during the S_B (first third of separation) period. Again, the time course of BT change, and recovery, was variable. Four infants had the lowest mean night BT value the S_1 night, one (20.2) on the S_3 night, and two (8.5 and 25.1) on the S_8 night.

Following reunion (the R_A day), 6 of the 7 infants had mean day BT values higher than any baseline mean day BT value. For the remainder of reunion (R_B), mean day BT values were similar to baseline. Mean night BT values for the group were higher than baseline for both R_A and R_B period. Examining BT values for the infants individually, mean night BT remained elevated (above the baseline mean) in 3 of 6 infants; in the other 4, values were similar to baseline.

c. *Sleep patterns.* For ease of discussion, we have divided the 12 different sleep variables into 3 groups. Group 1, generally having to do with wakefulness, consists of sleep latency, time awake during the night, number of arousals during the night, and total sleep time. Group 2, having to do with slow wave sleep, consists of Drowsy, Stage 2 and Stage 3–4 sleep. Group 3, having to do with REM sleep, consists of total REM time, number of REM periods, REM latency, inter-REM interval, and mean REM length.

The first group of variables is illustrated in Fig. 8. Sleep latency for the group the S_A night was increased to approximately 300% of baseline, with a 95% confidence interval well above baseline. During the remainder of separation, sleep latency was only very slightly increased above baseline. On the first night of reunion, sleep latency was less than baseline, and during the remainder of reunion had returned to closely approximate baseline values. Changes in mean values for time awake during the night followed a very similar pattern. The number of nocturnal arousals remained above baseline throughout all experimental periods including the first night of reunion, returning toward baseline values only during the last three days of reunion. In fact, the largest per cent increase in the number of arousals was during the last third of separation. Total sleep time (TST) was considerably diminished the first night of separation, and it remained somewhat lower than baseline for the remainder of separation, showing a slight increase during reunion. The changes in these four sleep variables during the S_A and S_B periods are quite similar to those described in our previous work for the first four days of maternal separation. It is noteworthy, however, that these group mean values do not return to baseline values throughout the longer 10-day separation.

Among the second group of variables (Fig. 9), there tended to be a slight decrease in Stages 3–4 sleep and Drowsy during separation, especially during the first night of separation. Except for a decrease the first night, Stage 2 sleep increased during

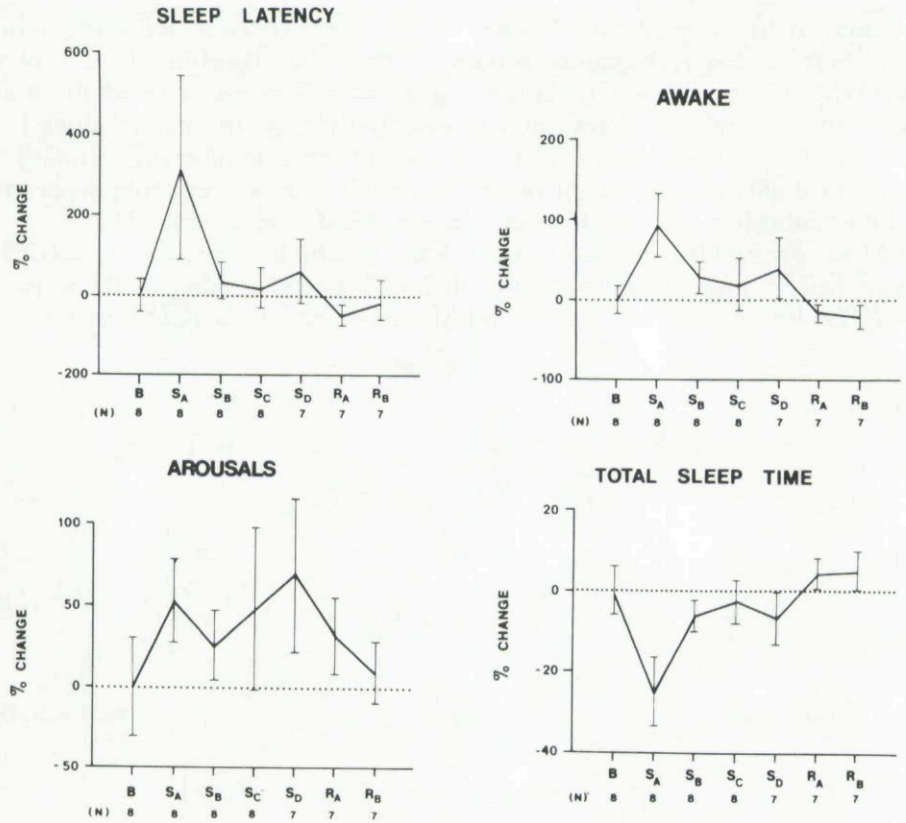


FIG. 8. Per cent changes (from baseline) in sleep latency, Awake time, Number of Arousals, and total sleep time during separation and reunion.

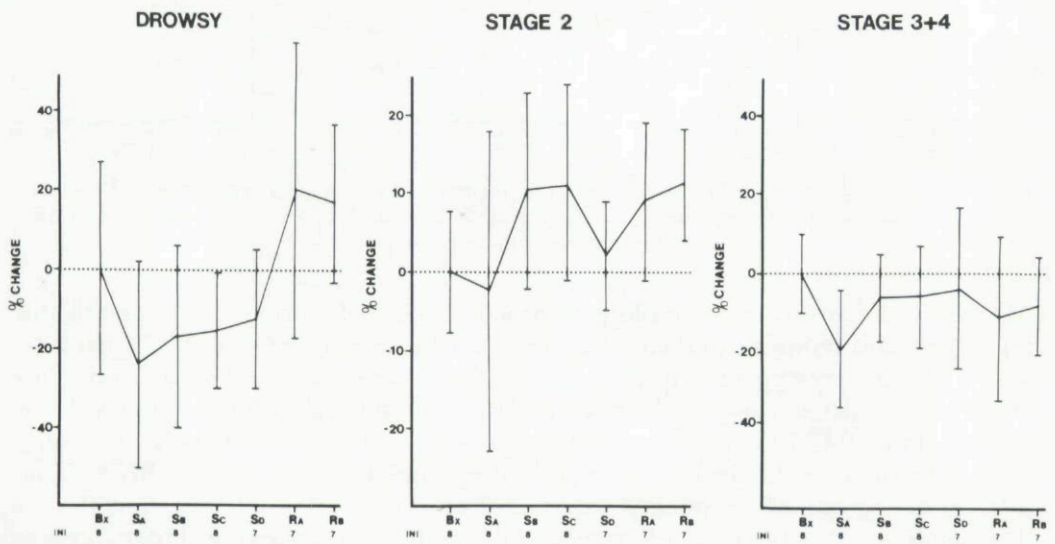


FIG. 9. Per cent changes (from baseline) in Drowsy, Stage 2, and Stages 3–4 sleep during separation and reunion.

separation. Both Stage 2 and Drowsy were above baseline following reunion, whereas Stage 3–4 sleep remained somewhat lower than baseline. Total slow wave sleep (SWS), the sum of Drowsy, Stage 2, and Stages 3–4 sleep, showed about a 10% decrease the S_A night, and returned to essentially baseline mean values for the remainder of separation, with evidence of a slight increase following reunion. This evidence indicates that the slight decrease in TST that accompanies separation is due almost entirely to the persistent decrease in REM sleep described below.

REM sleep variables are illustrated in Fig. 10. The first night of separation was accompanied by pronounced decreases in REM Time, number of REM periods, mean REM length, and increases in REM latency and inter-REM interval (IRI).

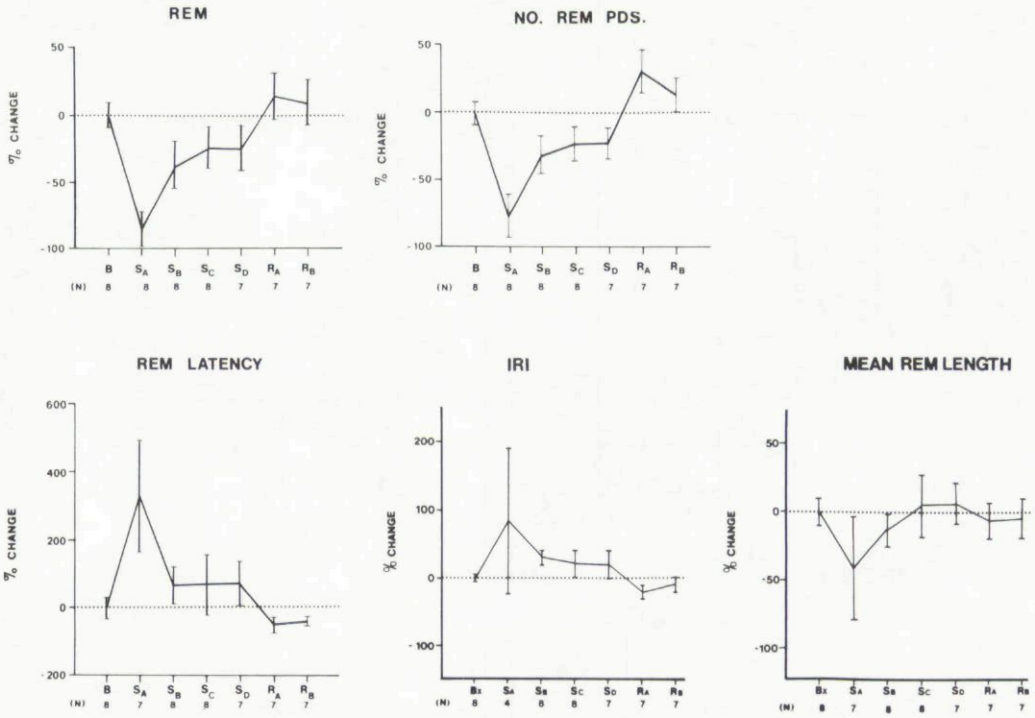


FIG. 10. Per cent changes (from baseline) in REM related variables during separation and reunion. One animal had no REM sleep the S_A night, and 3 had but one REM period, thus no IRI values could be computed for those subjects.

Mean REM period length rapidly returned to normal during the remainder of separation and reunion, whereas changes in REM time, number of REM periods, and REM latency appeared to persist throughout the duration of separation, with all three variables showing evidence of a relative rebound following reunion with the mother. Inter-REM interval has been shown to be a stable sleep variable (showing little individual variability) in normal monkey infants (Reite *et al.*, 1976). Time interval histograms of inter-REM intervals during separation were examined, and they suggested that the increase in mean IRI values accompanying separation was likely influenced primarily by occasional missed REM periods, but that there may

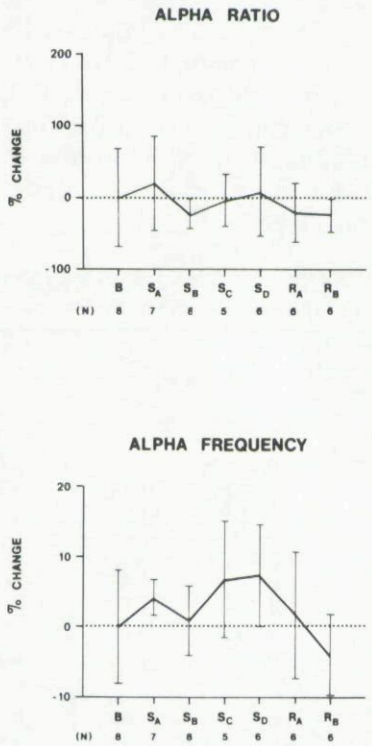


FIG. 11. Per cent changes (from baseline) in EEG alpha frequency and alpha ratio during separation and reunion.

be, in addition, a slight lengthening of the basic IRI during separation, which would suggest an alteration in the rhythmic control of REM sleep.

d. *Alpha frequency and ratio.* The first night of separation (S_A) alpha ratio and alpha frequency both tended to increase slightly. The effect on alpha frequency was more prominent with its associated decrease in variability (Fig. 11). During the early part of separation (S_B), which included the evolution of the depressive reaction, alpha frequency was essentially unchanged from baseline whereas alpha ratio mean values for the group diminished approximately 25%. Mean alpha ratio values returned towards baseline values during the S_C and S_D period, but there was a tendency for mean values to decrease again following reunion with the mother. Mean alpha frequency on the other hand, increased during the S_C and S_D periods, tending to return toward and eventually decreasing below baseline values during reunion. Again, considerable individual variability was the rule. For example, while the first separation night (S_A) was accompanied by slight increases in mean alpha ratio values for the group, in fact only 4 (of 7) animals showed increases, and 3 animals showed slight decreases.

3. *Physiology and behavior rankings across experimental conditions*

One of the questions we considered was whether the infants maintained their relative ranking across experimental conditions in terms of their physiological and

behavioral status. That is, for example, did infants with higher BT or HR mean values during the baseline period continue to have relatively higher values during separation and reunion (even though, as a group, HR and BT decreased). We computed Spearman rank order correlation coefficients between baseline and all separation and reunion conditions using absolute mean day and mean night HR and BT values, mean values for sleep variables, alpha ratio, and alpha frequency. These correlations are listed in Table 4.

TABLE 4. CORRELATIONS BETWEEN BASELINE AND EXPERIMENTAL CONDITION RANKINGS FOR PHYSIOLOGICAL VARIABLES

	S_A	S_B	S_C	S_D	R_A	R_B
HR and BT						
Mean night HR	0.45	0.38	0.60*	0.76†	0.66†	0.96†
Mean night BT	0.56*	0.96†	0.88†	0.89†	0.75†	0.77†
Mean day HR	0.90†	0.64†	0.24	0.26	0.10	0.32
Mean day BT	0.81†	0.82†	0.25	0.25	0.04	−0.03
Sleep Group 1						
Sleep Latency	0.10	0.45	0.36	0.07	0.41	0.61*
Awake	0.33	0.74†	0.69†	0.21	0.75†	0.11
No. Arousals	0.87†	0.88†	0.36	0.16	0.93†	0.86†
TST	0.00	0.81†	0.12	−0.25	0.34	−0.07
Sleep Group 2						
Drowsy	0.31	0.31	0.88†	0.71†	0.14	0.68†
St 2	0.02	−0.04	0.05	0.57*	0.46	0.79†
St 3–4	0.57*	0.81†	0.50*	0.64*	0.57*	0.43
SWS	0.05	0.69†	0.48	−0.32	−0.07	−0.21
Sleep Group 3						
REM	0.38	0.21	−0.21	−0.54	0.21	−0.04
No. REM Pds.	−0.12	−0.25	−0.48	−0.74†	−0.51	−0.11
$\bar{x}$ REM length	0.21	0.12	0.05	0.11	0.79†	0.46
REM latency	0.46	0.79†	0.64†	0.50	0.57*	0.61*
IRI	0.80*	0.21	−0.22	−0.25	0.32	0.67*
EEG Alpha						
Alpha frequency	0.96†	0.68†	0.56	0.72*	0.77†	0.93†
Alpha ratio	0.75†	0.86†	0.70*	0.26	0.51	0.71*

*Associated one-tailed probability of 0.1 or less.
†Associated one-tailed probability of 0.05 or less.

In the case of mean night HR, the infants did not maintain their rank between baseline and early to mid-separation, but by the latter third of separation, and through reunion, there was a much higher correlation between baseline and experimental condition rank. With mean night BT, the infants maintained their relative ranks over all conditions, although less well during S_A. Referring back to Fig. 6, we see that, for the group of 8 infants, mean night values continued to decrease from S_A to S_B, and then began to increase (recover). Mean night BT, on the other hand, was lowest during the first night of separation (S_A), and then began to recover. Thus, for both mean night HR and mean night BT, when evidence of recovery was noted, the infants showed a tendency to reassume their baseline ranking.

We know several infants had sustained changes (either increases or decreases) in HR during reunion, and (illustrated in Table 4) that the correlation between baseline mean night HR rank and reunion (R_B) mean night HR rank was quite high ($r = 0.96$); this implies that those infants who had sustained HR decreases were also the infants who initially had lower mean night HR values, and those infants who had sustained increases, had higher mean night HR values to begin with. This relationship was confirmed by a Spearman rank order correlation comparing baseline mean night HR values for the 8 infants with mean per cent change for the R_B condition ($r = 0.96$; $p = 0.001$).

In the case of mean day HR, a different pattern of relationships was found. Here, the infants maintained their baseline ranks for both S_A and S_B, but no longer. The S_A condition represents the agitation reaction, and infants thus maintain their ranks during agitation. Infants with lower baseline mean day HR values, however, had proportionately greater increases in mean day HR compared to infants with higher baseline mean day HR values ($r = -0.7381$, $p = 0.02$), a finding consonant with the Law of Initial Values (Wilder, 1950). Even though the baseline mean day HR rank order was not maintained past S_B, the overall mean day HR rank order was maintained from S_B through the remainder of separation and reunion (correlations not shown). Thus it appears that subsequent to the separation stress (and midway through separation), the infants had established a new rank order for mean day HR which was then maintained through reunion.

A very similar pattern of correlations was seen with mean day BT. Baseline rank order was maintained through S_A and S_B, following which a new rank order was established.

Among the Group 1 sleep variables (Table 4) Awake and No. of Arousals appeared to maintain rank to some degree, at least early in separation (S_B) and early in reunion (R_A). Among the Group 2 variables, Stage 3–4 sleep maintained rank the best, and among the Group 3 (REM) variables, only REM latency showed a tendency to maintain rank. Examining all correlations between all conditions, there was no evidence of a new and stable rank order being established for any sleep variable during the course of separation or reunion.

Both alpha frequency and alpha ratio showed a tendency to maintain their rank across experimental conditions, and there was no evidence for the establishment of new rank orders. Considering all physiological variables, the evidence for the establishment of new rank order patterns was found only for mean day HR and mean day BT.

As with physiology, we examined whether the infants maintained rank on behavioral variables across experimental conditions. Selected correlations are listed in Table 5. For many behaviors, scores on the R_A day were so low that the rankings are likely not meaningful. Infants tended to recover their baseline rankings on Activity Count and Motion Scores during the latter part of separation and reunion. This was not true for Play and Contact Other behaviors. The environmental contact behaviors Obx and Obs also displayed a recovery of baseline rankings following reunion, with Obs also maintaining rank during S_C and S_D. Of the 3 behaviors involving orality, at least one and often two exhibited maintenance of rank order during S_B, $S_C$and S_D, but oral Obx showed a transient reversal of rank order midway through separation (S_C). Baseline rankings were not well maintained for Away and

TABLE 5. CORRELATIONS BETWEEN BASELINE AND EXPERIMENTAL CONDITION RANKINGS FOR SELECTED BEHAVIORAL VARIABLES

	S_A	S_B	S_C	S_D	R_A	R_B
Activity Count	0.26	0.43	0.76†	0.86†	−0.18	0.62†
Motion	−0.24	0.33	0.57*	0.57†	−0.53*	0.38
Play	0.37	0.60*	0.19	0.32	—	0.00
Con Other	0.69*	0.29	−0.05	0.43	−0.05	0.40
Obx	0.40	0.33	−0.19	0.33	−0.41	0.74†
Obs	0.20	0.33	0.74†	0.62†	0.02	0.68†
Chew	−0.02	0.90†	0.45	0.62†	0.30	0.07
Self Orality	0.22	0.52*	0.93†	0.45	0.08	0.69†
Oral Obx	−0.26	0.40	−0.83*	−0.17	−0.46	0.21
Away	—	—	—	—	−0.30	0.26
Off	—	—	—	—	−0.25	0.43

*Associated one-tailed probability of 0.1 or less.

†Associated one-tailed probability of 0.05 or less.

Off behaviors following reunion. There was no evidence of the establishment of new rank order patterns for any of the behavioral variables listed.

4. *Correlations between early behavioral development and physiological response to separation*

An examination of separation induced changes in physiology as a function of early behavioral development provides one way of evaluating individual differences and provides information as to whether, by knowing something of an infant's early behavioral development, it can be predicted how that infant will respond physiologically to loss of the mother. Accordingly, we performed rank order correlations between behavior scores collected during months 2, 3 and 4 of life and later per cent changes in physiological variables during separation and reunion. Physiological variables used in these correlations included change in mean night HR, mean night BT, alpha frequency, alpha ratio, REM, REM latency, Awake, and No. of Arousals. Correlations between the change in each physiological variable for each of the six experimental periods and the behavioral variables for each 3 months were inspected. Table 6 presents those cases where there was a correlation with a one-tailed probability value of 0.1 or less for at least 2 of the 3 months; both correlation scores are listed.

We will not discuss this table in detail, but point out that since the physiological variables are per cent changes scores, the meaning of the correlations is influenced by the direction in which the physiological variables changed during separation. For example, mean night HR tended to decrease during S_A, S_B and S_C, therefore a negative correlation with measures of early behavioral development indicates a greater decrease, and such a relationship was found for Play and Punish behaviors during early behavioral development and change in mean night HR during the S_A period, indicating that the infants who played more and were punished more early in life tended to have a greater decrease in HR the first night of separation. During the R_A night, HR for most animals tended to increase above baseline values. Thus a negative correlation with early behavioral development would indicate less of an increase in HR the first night of reunion. Such a relationship was found between measures of Off, Away, and Activity level, suggesting that those animals who were active and off the mother more in early development had less of an increase in HR

TABLE 6. CORRELATIONS BETWEEN MEASURES OF EARLY BEHAVIOURAL DEVELOPMENT AND LATER CHANGES IN PHYSIOLOGY VARIABLES DURING SEPARATION AND REUNION. SEE TEXT FOR DETAILS

	S_A	S_B	S_C	S_D	R_A	R_B
Mean night HR % change	Play (−0.68, −0.71) Punish (−0.68, −0.60)	—	—	Away (−0.57, −0.62)	ActCt (−0.86, −0.55) Away (−0.74, −0.76) Off (−0.83, −0.64) PhyCon (0.83, 0.64)	—
Mean night BT % change	—	—	—	OLev (0.57, 0.61)	Play (−0.77, −0.75) Motion (−0.94, −0.77)	Play (−0.77, −0.94) Nip (0.77, 0.77)
Alpha freq. % Change	Off (−0.64, −0.78) Phy Con (0.64, 0.78) Nip (0.82, 0.82)	Obx (0.76, 0.63)	Nip (0.80, 0.80)	Off (−0.76, −0.77) Nip (0.77, 0.83) PhyCon (0.71, 0.77)	Play (0.67, 0.64)	—
Alpha ratio % Change	Nip (0.57, 0.57) IGro (0.79, 0.67) AGro (−0.61, −0.67)	Play (0.69, 0.71)	Obs (0.90, 0.70)	APla (0.82, 0.94) Obs (0.71, 0.77)	APla (0.70, 0.71) Obs (0.66, 0.71)	Away (0.71, 0.71) Motion (0.71, 0.89)
REM % Change	APla (0.71, 0.52)	—	Off (0.52, 0.83)	Act (0.64, 0.68)	RGro (0.68, 0.93)	Act (0.61. 0.96) Rest (−0.57, −0.61) Motion (0.57, 0.57) Apart (0.88, 0.64) OLev (0.64, 0.82)
REM latency % Change	Wean (−0.91, −0.77) Motion (0.75, 0.96) Apart (0.85, 0.61) Away (0.93, 0.86) OraSelf (−0.61, −0.71) OLev (0.93, 0.79)	—	RGro (−0.71, −0.52) Obs (−0.71, −0.65)	Away (0.57, 0.82) OLev (0.86, 0.89) Play (0.61, 0.65) Motion (0.82, 0.93) Wean (−0.85, 0.61) OraObx (0.71, 0.82)	—	Off (0.68, 0.68) PhyCon (−0.68, −0.68) RGro (0.79, 0.68)
Awake % Change	Act (0.59, 0.60) Carry (−0.55, 0.60) Play (−0.54, −0.75)	Punish (−0.78, −0.74)	—	Rest (−0.57, −0.93) OLev (0.68, 0.64)	Off (0.64, 0.68)	—
No. arousals % Change	Apart (0.57, 0.63) Away (0.55, 0.62) OLev (0.64, 0.62)	Act (−0.57, −0.79) Play (0.85, 0.69)	Punish (0.68, 0.71) Play (0.71, 0.61)	OLev (0.71, 0.64) Away (0.71, 0.68)	Act (−0.79, −0.61) Off (−0.71, −0.71) PhyCon (0.71, 0.71) OraObx (−0.61, −0.61) Chew (−0.64, −0.64) Nip (0.75, 0.75) OLev (−0.64, −0.64) IGro (0.76, 0.76)	Act (0.93, 0.57) Nip (−0.64, −0.57) OLev (0.68, 0.61) PhyCon (−0.89, −0.71) Off (0.89, 0.71)

the first night of reunion. The other correlations should be interpreted in a similar manner.

While the patterns of correlations are far from clear cut, all in all they do suggest that infants who are more active (and also therefore away from the mother more) in months 2, 3, and 4 of life have the greatest changes in HR and several sleep variables early in separation, and also tend to be the ones with greater REM rebounds following reunion.

Hinde and Spencer-Booth (1970) have used $\%A-\%L$ scores as one measure of 'tension' in the mother–infant relationship, with high scores indicating that the infant (as opposed to the mother) was primarily responsible for maintaining proximity. To assess whether this type of 'tension' early in the infant's life would influence the nature of the mother–infant reunion following separation, we correlated the $\%A-\%L$ scores for infants during months 2, 3, and 4 of life, with change scores for Off, Away, Other Level, REM Sleep and Mean Night HR for the R_A and R_B period. These correlations are presented in Table 7. All 3 of the behavioral variables (Off, Away, and Other Level) decreased to values below baseline during R_A. The negative correlations between these change scores and $\%A-\%L$ for months 2, 3, and 4 indicate that those animals whose Off, Away, and Other Level scores decreased the most were those who had spent more time maintaining proximity to the mother. However, a similar pattern was not seen during R_B where the correlations, although generally tending to be positive, were not large. The change scores for both REM sleep and Mean Night HR during R_A correlated positively with $\%A-\%L$ scores. Since both REM and Mean Night HR tended to increase the R_A night, this suggested that the increases were more prominent in infants who had been primarily responsible for maintaining proximity during early development and whose relationship with the mother, on this basis, may have had a greater degree of 'tension'. Once again, this pattern of correlation disappeared during R_B.

TABLE 7. CORRELATIONS BETWEEN $\%A-\%L$ SCORES FOR MONKEYS 2, 3 AND 4, AND % CHANGE SCORES FOR SEVERAL BEHAVIORAL AND PHYSIOLOGICAL VARIABLES DURING R_A AND R_B

		Mo. 2	Mo. 3	Mo. 4
% Change Scores R_A	Off	−0.28	−0.45	−0.63†
	Away	−0.28	−0.45	−0.63†
	Other Level	−0.37	−0.49	−0.55*
	REM	0.58	0.47	0.64*
	HR	−0.15	0.52*	0.10
% Change Scores R_B	Off	0.04	0.38	0.33
	Away	−0.26	0.25	0.31
	Other Level	0.19	−0.10	0.10
	REM	−0.35	0.20	0.36
	HR	−0.23	0.61*	0.50

*Associated one-tailed probability 0.1 or less.
†Associated one-tailed probability 0.05 or less.

DISCUSSION

1. *Behavior*

The literature on maternal separation in monkey infants demonstrates quite clearly that the nature of the behavioral reaction is very sensitive to the characteristics of the environment in which the separation experiments are carried out (Chappel and Meier, 1975; Hinde and McGinnis, 1977; Kaufmann and Stynes, 1978; Mineka and Suomi, 1978). This being the case, we will attempt to compare our current results with findings obtained from studies using a similar social group setting.

Separation studies similar to ours have been carried out by Kaufman and his collaborators using both pigtailed and bonnet macaques (Kaufman and Rosenblum, 1967a; Rosenblum and Kaufman, 1978; Kaufman and Stynes, 1978), and extensive studies in rhesus monkeys have been performed in Hinde's laboratories (Hinde and Spencer-Booth, 1970, 1971; Spencer-Booth and Hinde, 1971a, 1971b; Hinde, 1977; Hinde and McGinnis, 1977). Both pigtailed and rhesus infants exhibit a characteristic two-phased reaction with a period of agitation, characterized by increased locomotor behavior, usually of a searching nature, and increased vocalization, followed by a period of behavioral depression, accompanied by a slouched posture, decreased activity with slowing of movement, a marked diminution of play behavior, and in the case of Hinde *et al.*'s infants, an increase in eating and contact with other animals. Bonnet infants do not normally display a profound behavioral depression when the mother is removed while they remain in a bonnet social group. In a mixed species (bonnets and pigtailed monkeys) group, however, where both the mother and other conspecifics are removed, and the separated bonnet infant remains with the pigtailed group members, the behavioral depressive reaction is very similar to that seen in pigtailed infants (Kaufman and Stynes, 1978). The behavioral reaction during separation in our infants is therefore quite similar to the separation induced behavioral changes described by other investigators. Hinde and Spencer-Booth (1971) found no substantive evidence for behavioral recovery during 13-day maternal separations in rhesus infants; in our infants evidence of behavioral recovery during the S_D period was minimal at best with small decreases in Slouch scores and increases in Play scores. During reunion we found evidence for relatively transitory alterations in the mother–infant relationship that were most pronounced the day of reunion (R_A). Longer lasting changes as described by Spencer-Booth and Hinde (1971c) were not observed. It is not clear whether this may be due to different social group structures, species differences, or differences in observational and/or analytical methodologies.

2. *Physiology*

The changes in HR, BT, and sleep patterns we have described in this paper during the early S_A and S_B periods are similar to changes we previously found during shorter (4 day) separations (Reite *et al.*, 1978; Reite and Short, 1978). It is important to note that these physiological changes are not trivial. They are not, for example, HR changes of a few bpm lasting for a few minutes. They are major alterations in physiological functions sustained over many days. We also now have evidence for some measure of recovery of certain physiological functions during the longer period of separation, as both BT and HR group mean values were approaching baseline

values during the S_D period (although again exhibiting marked individual variability) and we have additional evidence for long term changes in HR in some infants. Both R23.1 and R9.3 had persistent decreases in HR throughout 4 days of reunion. When R23.1 was recorded again 1 month later, HR had returned to baseline levels. In the case of R9.3, physiological recording continued for 15 days following reunion with the mother, and HR remained low throughout this period. Further, R9.3 was studied again 12 months later (using a new telemetry transmitter), at which time HR was still well below the original baseline values. Whether the findings 12 months later represent a persistent separation-induced bradycardia, or a normal maturational decrease in HR is as yet unclear.

Sleep pattern changes were for the most part more resistant to recovery, with REM latency, REM time, and number of REM periods apparently showing a plateau effect during the S_C and S_D periods rather than continuing the trend started between S_A and S_C. Sleep latency, Number of Arousals and time Awake all increased from S_C to S_D. Also of note in the present data is the evidence of a REM rebound during reunion. Group mean values for REM time and Number of REM periods were increased during reunion, and IRI and REM latency were decreased. These changes are indicative of a REM rebound phenomenon in this group of infants, an effect not noted following 4-day maternal separations (Reite and Short, 1978).

Sleep pattern changes may result from at least two major influences. The initial effects are likely a composite of both a first night effect (Agnew, Webb and Williams, 1966) as well as the more specific disruption due to the separation experience. As the infant becomes more accustomed to the change in sleep environment, the 'first night' type of effect should be minimized, and we should be left with the residual due primarily to the separation *per se*. We previously found in three surrogate-reared infants that sleep pattern changes (which we attributed primarily to a first night effect) had returned to within one standard deviation of baseline values by the 4th night of separation (Reite, Short and Seiler, 1978). In the present study we believe those sleep pattern changes continuing into the S_C and S_D period are likely due to primarily to more specific effects related to the disruption of an attachment bond. In general, such effects appear confined primarily to arousal and REM systems, and do not appear to significantly influence SWS systems except insofar as they are disrupted or altered by increases in Sleep Latency and Number of Arousals.

The fact that alpha ratio decreased (with alpha frequency remaining constant) during the early part of separation (S_B), suggests that there may have been an alteration in EEG regulation with the admixture of greater amounts of lower frequency activity. Preliminary studies of high-central EEG theta activity, thought to be a correlate of phasic motor inhibition (Reite *et al.*, 1974a) have suggested a slowing of frequency during the early part of the depressive reaction (Short, Iwata and Reite, 1977; Reite *et al.*, unpublished data).

As we examined the overall pattern of changes in the physiology of these infants during the period of maternal separation, there appear to be two distinct effects. First, there is a general tendency (for the group) for physiological variables to change in a patterned manner. For example, we most often observe decreases in heart rate and body temperature and several REM sleep measures and increases in time Awake, Number of Arousals and REM latency. This general tendency is

illustrated in our group mean scores. There is also, however, a very pronounced increase in intra-individual variability during the period of separation. This gives the distinct impression that the overall homeostatic regulation of physiological functioning is impaired. To what extent these two apparently different types of effects are mediated by different mechanisms, or may represent the manifestation of similar underlying mechanisms, we do not yet know. What is apparent is that the animal's capacity to regulate its own physiological status is considerably impaired by having experienced loss of the mother.

3. *Correlations between early behavioral development and later response to separation*

In previous work from Hinde's laboratory, it was found that in rhesus infants, those infants who were more distressed on reunion were those who prior to separation had exhibited more 'tension' in the mother–infant relationship, as manifested by being primarily responsible for maintaining proximity to the mother, and having had a greater proportion of attempted nipple contacts rejected (Hinde and Spencer-Booth, 1971; Spencer-Booth and Hinde, 1971b; Hinde and McGinnis, 1977). The data from our infants (Table 7) indicate that infants with higher $\%A-\%L$ scores in early development, and therefore possibly with more 'tension' in the mother–infant relationship, also tended to spend less time off and away from the mother the day of reunion, as well as having greater increases in mean night HR and REM sleep the first night of reunion. In both cases the relationship appeared to be short-lived however.

In our previous work on behavioral development and physiological status of normal mother-reared group living pigtailed infants, we found that early measures of behavior generally correlated quite well with later behavioral development and subsequent physiological status (Reite and Short, 1980). Infants were found to vary individually with respect to behavioral measures such as general level of activity and with respect to certain aspects of the mother–infant relationship. These differences appeared early and persisted. We interpreted our findings (in a qualified manner) as compatible with the general concept of developmental continuity. The infants described in this study on maternal separation were also included in the previous study on normal development.

In the present study, it appears that we have reached the position where we can predict with a fair degree of certainty how a group of pigtailed infants, studied in a similar social group living experimental paradigm, will react to loss of the mother both behaviorally and physiologically. Individual variability is high, however, and overall, the correlations between various measures of early behavioral development and the subsequent response to separation, while suggestive, is not clear cut, suggesting that even knowing a good bit about a given infant's early behavioral development, it is difficult to predict that infant's physiological reaction to separation, especially early in separation. In other words, the behavioral developmental measures we have do not appear to be very good predictors of the nature of the physiological reaction to the disruption of an attachment bond. It would appear therefore that the introduction of this particular stress (the disruption of an attachment bond) may tend to displace the organism from its own "epigenetic trajectory" (as used by Waddington, 1968) in an essentially unpredictable manner. Whether

this displacement is an enduring one, with long term developmental changes, awaits further research. We have found evidence for relatively long lasting alterations in HR in several infants, and evidence of separation-induced behavioral changes lasting up to 2 years have been described (Spencer-Booth and Hinde, 1971c).

4. *Mechanisms underlying physiological changes*

The initial increases in both HR and BT that occurred immediately following separation were greater than could be accounted for by the increased activity alone, as they occurred very rapidly (immediately in the case of HR, within minutes in the case of BT) after separation, and they were of greater magnitude (especially as far as BT was concerned) than would have been seen during a similar period of active play. These increases occurred even though the infant was not actually running vigorously about all the time. Interspersed periods of walking, running, and sitting, while intently scanning the environment, are behaviors which more closely characterized the aroused and tense infant. We expect therefore that the HR and BT increases during the agitation period were due to a massive activation of the sympathetic adrenergic system. Indeed, in a previous study, Breese *et al.* (1973) found evidence of a neurally mediated stimulation of adrenal catecholamine synthesis in infant rhesus monkeys who had been separated from their mothers for 6 days, but who, according to these investigators, were still exhibiting the protest or agitation portion of the behavioral reaction.

Essentially opposite changes in HR and BT were observed during the subsequent depressive reaction. We believe the overall weight of the evidence would tend to again implicate autonomic mechanisms. First and most obvious, a marked increase in vagal tone would certainly help explain the bradycardia and cardiac arrhythmias found during the period of behavioral depression in most infants. All such interpretations are limited, of course, by the fact that we only have data for HR, and not blood pressure, dP/dt, or an accurate clinically relevant assessment of the nature of the arrhythmias. The alleged importance of vagally mediated effects on heart rate and rhythm have been pointed out by Richter (1957) who stated:

> "A phenomenon of sudden death has been described that occurs in man, rats, and many other animals apparently as a result of hopelessness; this seems to involve overactivity primarily of the parasympathetic system."

Alternatively, or perhaps additionally, there may be a diminution in central beta-adrenergic tone. Obrist *et al.* (1978) found that beta-adrenergic influences on HR (and other measures) were greater (in human subjects) when subjects either had, or believed they had, some control over an avoidance task. On the other hand, beta-adrenergic influences were either minimal or rapidly dissipated under conditions where no control was possible. The separated and depressed monkey infant, possibly having perceived its state as one of "helplessness" (Seligman, 1975), may experience a net decrease in central beta-adrenergic activity. We do not yet know however, to what extent beta-adrenergic activity influences HR in normal pigtailed infants.

Other catecholamine related mechanisms have been implicated in the reaction to separation in monkey infants based upon behavioral observations. Morrison, Kraemer and McKinney (1976) found that AMPT, a drug which blocks catecholamine synthesis, potentiated the protest–despair response to peer separation in 18 month old rhesus monkeys, and Suomi *et al.* (1978) found that imipramine treat-

ment (which blocks catecholamine reuptake at the presynaptic terminal) results in behavioral improvement in rhesus infants subjected to repetitive peer separations.

The decreases in HR, and possibly BT as well, are suggestive of a decrease in energy expenditures and possibly oxygen consumption as well, although we have no direct evidence for this. In a study of energy expenditure in school age boys, Bradfield *et al.* (1971) found that 87% of the variation (in kilocalories per minute) could be accounted for by differences in heart reate, and Berg (1975) also found HR to be a good indicator of energy expenditure in children with cerebral palsy. The older EEG literature also provides examples of EEG (alpha) frequencies varying directly as a function of body temperature (Hoagland, 1936), and metabolic rate (Rubin, Cohen and Hoagland, 1937). Our data is consonant with such evidence for a decrease in metabolic activity and thus also supports those investigators who conceptualize the depressive component of agitation–depression as a manifestation of a more basic conservation–withdrawal reaction (Kaufman and Rosenblum, 1967b; Engel and Schmale, 1972; Schmale, 1973).

Let us introduce a caveat at this point. We have suggested several possible mechanisms underlying the physiological changes accompanying maternal separation in the monkey infant from a somewhat biased and essentially narrow viewpoint. It is that viewpoint provided by the physiological measurements we have access to, and the methods we used to analyze those measurements. It neglects to consider many other possible physiological changes that may also be present, and it neglects to consider the important area of hormonal changes that may also accompany the agitation–depression reaction. We do not yet possess the data necessary to conceptualize the issue from a holistic or integrative viewpoint, the ultimate importance of which has been elegantly summarized by John Mason (1968). We believe that our overall approach does provide a step in that direction, however, by providing the opportunity to integrate complex behaviors with multi-variable physiology in the unrestrained subject.

While we may not yet understand the mechanisms that underly the physiological reaction to separation in the monkey infant, we do have specific evidence that it is due not so much to the loss of mothering, as to the loss of a specific mother. One pigtailed infant (not included among the present group) was adopted by another childless female shortly after separation, but, nonetheless, exhibited many behavioral and physiological signs of depression, including decreases in play behavior, slouched posture, sleep disturbances, and decreases in body temperature (Reite *et al.*, 1978). This evidence suggests that the critical factor in the etiology of the reaction is the loss of a specific mother, that is, the disruption of an attachment bond.

CONCLUSIONS

These data demonstrate that separations or losses may be accompanied by a complex series of physiological changes which underscore the importance of attachment bonds, and their disruption, to the integrative functioning of high primates. Conceptually such observations suggest important relationships between attachment behavior and its neurobiological subtrates, and the regulation of affective behavior, and may ultimately contribute to our elucidation of the pathophysiology of grief (Reite, 1979).

From a clinical standpoint, such observations reflect on the necessity of viewing losses and separations from a more comprehensive viewpoint than that provided by behavioral change or the imputed function of psychic or other psychological mechanisms alone. An appreciation of the broader implications of such phenomena should include anticipation of accompanying physiological changes, with alterations in autonomic homeostasis as one facet. Such relationships may be of special importance when dealing with losses or separations in human infants and children.

SUMMARY

The influence of 10 days of maternal separation on behavior and physiology was studied in 8 social group living pigtailed macaque (*M. nemestrina*) monkey infants. Maternal separation was accompanied by behavioral changes which included slowing of movement, decreases in play, increases in oral behaviors, and assumption of a characteristic slouched posture and sad facial expression. Physiological changes included decreases in heart rate and body temperature, sleep disturbances that were most pronounced in the Awake and REM sleep systems, and changes in EEG alpha power-spectra. Individual variability in the behavioral and physiological response to maternal separation was marked, and showed only a minimal correlation with measures of early behavioral development and the early mother–infant interaction. Some physiological changes appeared to be relatively long lasting.

The data demonstrate that the disruption of an attachment bond may be accompanied by pronounced physiological changes suggestive of a general impairment of autonomic homeostatic regulatory processes. The findings have important implications for our understanding of the pathophysiology of grief, and the physiological concomitants of separation, loss, and depression in children.

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